

VASILEIOS BERTSOS

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Education

University of Ioannina

October 2023 – Present

Diploma in Engineering (Integrated Master), Computer Science and Engineering

Ioannina, Greece

Relevant Coursework

- Data Structures
- Algorithmic Analysis
- Object-Oriented Programming
- Artificial Intelligence

Technical Skills

Languages: Java, Python, C, MATLAB, HTML.

Artificial Intelligence: Search Algorithms (A*, UCS), Metaheuristic Optimization (Simulated Annealing), Constraint Satisfaction Problems (CSPs), Generative AI (OpenAI API)

Hardware & Prototyping: 3D Printing (FDM/Resin), 3D Modeling (Blender), Hardware Integration, Digital Logic Design.

Tools: Git, Linux, VS Code, Quartus.

Experience

Computer Science & Engineering Dept, UoI

March 2025 – Present

IT Support Engineer (Part-time)

Ioannina, Greece

- Provide software troubleshooting and IT support for students and faculty.
- Assist in systems administration and maintaining the operational efficiency of university labs.
- Co-organized and led a 6-session hands-on workshop on PC Assembly and system architecture, training 150+ undergraduate and postgraduate students.
- Mentored participants in hardware compatibility, physical integration (CPU, RAM, storage), and practical troubleshooting.

Projects

Sign Language Image Classifier | Python, Scikit-learn, Pandas, NumPy

April 2026

- Developed an ML classification pipeline to recognize American Sign Language (ASL) from 27,000+ images, performing pixel normalization and stratification.
- Trained and evaluated multiple classification models (k-Nearest Neighbors, Support Vector Machines, Decision Trees), optimizing hyperparameters to maximize performance.
- Achieved exceptional classification accuracy, validated via Accuracy and F1-score metrics.
- Implemented a custom prediction algorithm to successfully decode and reconstruct hidden text phrases from sequences of unseen test images.

AI Pathfinding Agent (8-Puzzle Variant) | Java

March 2026

- Co-developed an AI agent within a student team to solve a complex state-space problem.
- Implemented UCS and A* search, designing a mathematically proven admissible heuristic that significantly reduced the algorithm's search space and execution time.

Sudoku Solver (Simulated Annealing) | Java

March 2026

- Modeled the 9x9 Sudoku as a Constraint Satisfaction Problem (CSP).
- Collaborated to build an automated solver using the Simulated Annealing algorithm to minimize conflicts.

CanSat in Greece Competitor — Team ZosiMinds

2022 & 2023

- National finalist in the ESA-backed 'CanSat in Greece' competition.
- Co-designed and constructed the physical structure of a functional mini-satellite.
- Applied hands-on hardware integration and mechanical assembly under strict deadlines.

Languages

Greek: Native

English: Full Professional Proficiency (C2 Proficiency)

Certifications

Prompt Engineering & Programming with OpenAI

July 2025

Columbia+

- Integrating Large Language Models (LLMs) into real-world applications.